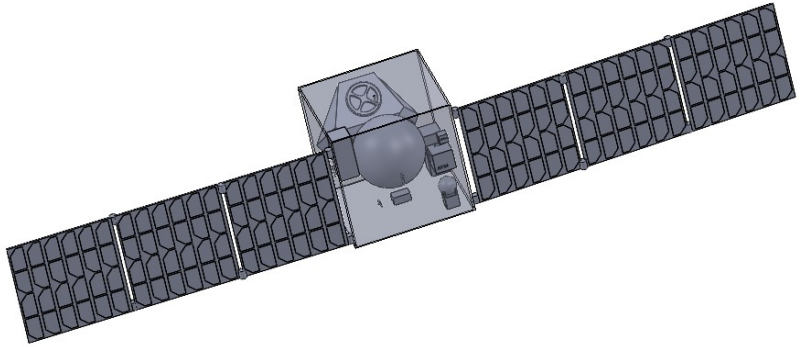
ChLOE



Chasers in Low Orbit around Earth

Zaara Amin, Caden Akins, Ruelen Sarion, Saahil Sinha, Matthew Silva, Alba Borrego Pablos, Ignacio García Pereña, Jorge Sanz, Fotini Ioannides, Anthony Sinigiani

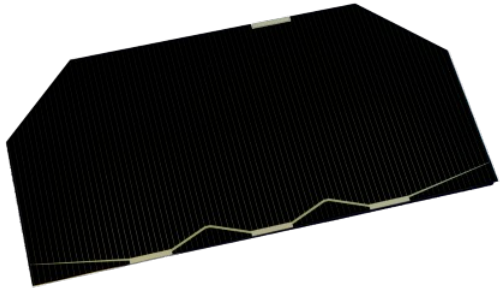
UCLA Samueli School of Engineering

Power

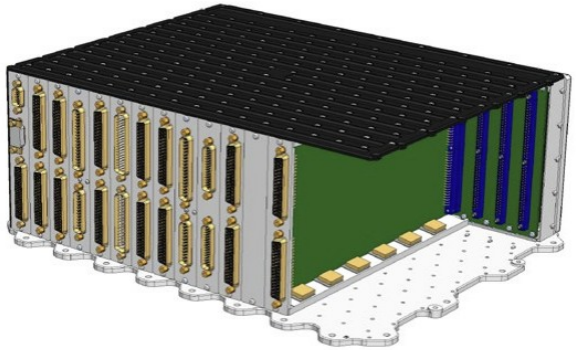
Power – EPS Architecture

Solar Panels to generate. Battery to Store. PCDU to distribute.

GENERATE	STORE	DISTRIBUTE
Azur Space 3G30C-Advanced	Saft VES16 (NCA Li-ion)	Terma PCDU (S3R topology)







Power – EPS Architecture

Solar Panels to generate. Battery to Store. PCDU to distribute.

GENERATE	STORE	DISTRIBUTE
Azur Space 3G30C-Advanced	**Saft VES16 (NCA Li-ion)**	**Terma PCDU (S3R topology)**
BOL efficiency: $\eta = 29.5\%$	Specific energy: >155 Wh/kg	Bus voltage: 100 V (selected)
EOL factor: 0.919 (1×10^{15} e^-/cm^2) (Worst case)	Depth of discharge: 30%	Bus current at EOL: 48.4 A
Configuration: 2 wings × 7.29 m^2	Design capacity: 1,267 Wh	PCDU efficiency: $\eta = 96\%$
Total area: 14.58 m^2	EOL fade factor: 0.95 (5 year design life)	Conversion losses: 193.5 W
BOL power: 5,264 W	Mission cycles: 27,518	PCDU mass: 5 kg estimated
EOL power: 4,838 W	Rated cycles: >60,000	Harness mass: 15 kg estimated
Mass: 57.5 kg	Mass: 8.2 kg	Mass: 20 kg

Power – Solar Array Sizing

Methodology & Key Equations

Orbital period

$$T = 2\pi\sqrt{\frac{r^3}{\mu}}$$

r = R_e + h = 6,921 km

95.5 min

Eclipse duration

$$t_{ecl} = \left(\frac{T}{\pi}\right)\arcsin\left(\frac{R_e}{r}\right)$$

Worst case β = 0°

35.6 min

Required array power

$$P_{sa} = \frac{\left(\frac{P_e T_e}{X_e} + \frac{P_d T_d}{X_d}\right)}{T_d}$$

SMAD Eq. 11-5

4,837.7 W

EOL output per m²

$$P_{EOL/m^2} = S \cdot \eta_{BOL} \cdot packing \cdot EOL_{factor}$$

1360 × 29.5% × 90% × 0.919

331.8 W/m²

Minimum array area

$$A = \frac{P_{sa}}{P_{EOL/m^2}}$$

14.58 m²

Input Parameters & Results

Parameter	Value
Solar irradiance S	1,360 W/m²
Cell BOL efficiency η_BOL	29.5% (Azur Space 3G30C-Adv.)
EOL degradation factor	0.919 (1×10¹⁵ e⁻/cm²)
Packing factor	90%
Sunlit path efficiency Xe	0.85 (MPPT + line + diode)
Eclipse path efficiency Xd	0.96 (S3R PCDU)
Eclipse power P_e	1,236 W (MEV − PCDU + 30%)
Thrusting power P_a	3,463 W (MEV − PCDU + 30%)
Required P_sa	**4,837.7 W**
Array area	**14.58 m²**
Configuration	**2 wings × 7.29 m²**
BOL / EOL power	**5,796 W / 4,837.7 W**
Array total mass	**57.5 kg**

Power – Battery, PCDU & Mass

	Solar Array	Battery	PCDU
Component	Azur Space 3G30C-Adv.	Saft VES16	Terma PCDU
Key output	EOL: 4,838 W	1,267 Wh | 30% DoD	100V | η=96%

EPS Mass Breakdown		
Component	**Breakdown**	**Mass (kg)**
Solar Array	Cells 25.7 + Structure 21.9 + Deploy 10	57.5
Battery	Saft VES16 × 1,267 Wh at 155 Wh/kg	8.2
PCDU + Harness	5 kg PCDU + 15 kg harness	20.0
TOTAL EPS MASS	Solar + Battery + PCDU + Harness	**85.7**

Power Summary	
Parameter	**Value**
Eclipse duration	35.6 min (β = 0° worst case)
Mission life	5 yr design life (~1 yr mission) (27,518 eclipse cycles)
Solar driver (Thrusting)	3,463 W
Battery driver (Eclipse)	1,236 W

PEL Overview

PEL Overview - Power Equipment List

Subsystem	Key Components	CBE (W)	Contingency	MEV (W)
Imaging Payload	Mast-Z Camera + CFC-400 Flight Computer	42.4	10%	46.6
Refueling Payload	Heaters, Valves, Sensors, Control Electronics	95.0	15,3%	109.5
Propulsion (Hall+PPU)	SPT-100 Hall Thruster + Power Processing Unit	1,485.0	10%	1,633.5
ADCS	Reaction Wheels ×4, RCS ×72 components, Gyros, Trackers	901.8	10%	992.0
Telecom / C&DH	SCR-106 Transmitter, TT&C Rx, Flight Computer	11.4	17,5%	13.4
Thermal	Survival Heaters (511.1 W, eclipse only) + Controllers	521.1	29,6%	675.4
EPS (PCDU losses)	Power Control & Distribution Unit — 4% of bus power	193.5	10%	212.9

Power Totals (W)	Standby	Nominal	Imaging	Refuell	Thrusting	Rendezvous	Downlink	Eclipse	Safe Mode
Total CBE	449.2	483.5	1162.0	1204.6	2614.6	1129.6	483.6	964.6	434.2
Total MEV	494.3	532.5	1278.8	1330.7	2876.7	1243.2	532.6	1163.9	477.8
MEV + 30% margin	642.6	692.2	1662.5	1729.9	3739.7	1616.2	692.4	1513.0	621.2

PEL Overview - Power Equipment List

Component / Mode	Standby	Nominal	Imaging	Refueling	Thrusting	Rendezvous	Downlink	Eclipse	Safe
Mast-Z Camera (active)	–	–	●	–	–	–	–	–	–
Mast-Z Camera (standby)	●	●	–	●	●	●	●	●	–
CFC-400 Flight Computer	●	●	●	●	●	●	●	●	–
Refueling System	–	–	–	●	–	–	–	–	–
SPT-100 + PPU (1,485 W CBE)	–	–	–	–	●	–	–	–	–
ADCS (RW always on + RCS)	●	●	●	●	●	●	●	●	●
Telecom + C&DH	●	●	●	●	●	●	●	●	●
Survival Heaters (511 W)	–	–	–	–	–	–	–	●	–
Thermal Ctrl / Sensors	●	●	●	●	●	●	●	●	●
PCDU (losses, all modes)	●	●	●	●	●	●	●	●	●
Total CBE	449.2	483.5	1162.0	1204.6	2614.6	1129.6	483.6	964.6	434.2
Total MEV	494.3	532.5	1278.8	1330.7	2876.7	1243.2	532.6	1163.9	477.8
MEV + 30% margin	642.6	692.2	1662.5	1729.9	3739.7	1616.2	692.4	1513.0	621.2